NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 11

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

Which of the following Metasploit modules is used to take advantage of a known vulnerability on a target system?

a. Payload
b. Exploit
c. Encoder
d. Auxiliary

Correct Answer: b

Detailed Solution: The exploit module is specifically designed to take advantage of a vulnerability in a target system or application. The payload establishes a communication channel after exploitation, the encoder obfuscates payloads to evade detection, and the auxiliary module performs scanning or brute-force tasks.
Thus, the correct option is (b).

QUESTION 2:

Which command is used to launch the Metasploit Framework console?

a. msfvenom
b. msfconsole
c. metasploit-cli
d. msf-start

Correct Answer: b

Detailed Solution:

The “msfcnsole” command is used to launch the Metasploit Framework.
Thus, the correct option is (b).

QUESTION 3:

Which Metasploit command is used to check which target operating systems are compatible with a selected exploit?

a. show targets
b. set targets
c. check compatibility
d. show exploits
e. show payloads

Correct Answer: a

Detailed Solution: The "show targets" command displays the list of compatible target operating systems for the currently selected exploit module, helping the user select the correct target environment.
Thus, the correct option is (a).

QUESTION 4:

Which Metasploit option is used to specify the target's IP address (the remote host to be attacked)?

a. set LHOST
b. set LPORT
c. set RHOST
d. set RPORT

Correct Answer: c

Detailed Solution: RHOST is used to set the remote host (target) IP address, while LHOST sets the attacker's (local) IP, and RPORT/LPORT set the respective port numbers.
Thus, the correct option is (c).

QUESTION 5:

Which command in Metasploit's Meterpreter shell is used to retrieve stored user account password hashes from the target system?

a. Getuid
b. Getsystem
c. Hashdump
d. Sysinfo
e. ps

Correct Answer: c

Detailed Solution: The hashdump command retrieves stored password hashes from the target system's SAM database (on Windows) or equivalent, which can then be used for offline password cracking. getuid shows the current user ID, getsystem attempts privilege escalation, sysinfo shows target system details, and ps lists running processes.
Thus, the correct option is (c).

QUESTION 6:

Which tool is specifically designed to perform brute-force discovery of hidden files and directories on a web server?

a. SQLMAP
b. Dirb
c. Hydra
d. Crunch
e. Hashcat

Correct Answer: b

Detailed Solution: Dirb performs brute-force discovery of existing and hidden pages or directories on a web server using a wordlist. SQLMap automates SQL injection attacks, Hydra and Crunch are password-cracking and wordlist-generating tools, respectively; and Hashcat cracks password hashes.

Thus, the correct option is (b).

QUESTION 7:

If a web page is vulnerable to error-based SQL injection, what is the expected behavior upon incorrect/malicious user input?

a. It will display a database error message revealing information about the query structure.
b. It will silently return no output at all.
c. It will redirect the user to a login page automatically.
d. It will crash the entire web server permanently.

Correct Answer: a

Detailed Solution: In error-based SQL injections, malformed or malicious input causes the application to generate a visible database error message, which can leak useful information about the underlying query structure and database schema.

Thus, the correct option is (a).

QUESTION 8:

Which SQLMAP option is used to list all database names available on the target system?

a. --tables
b. --dbs
c. --dump
d. --current-db
e. --passwords

Correct Answer: b

Detailed Solution: The --dbs option lists all available database names on the target. --tables lists tables within a chosen database, --dump extracts data from specified tables, --current-db shows the currently active database, and --passwords retrieves password hashes.

Thus, the correct option is (b).

QUESTION 9:

Which of the following is/are recognized types of Cross-Site Scripting (XSS) attacks?

a. Stored XSS
b. Reflected XSS
c. DOM-based XSS
d. SQL-based XSS
e. Cookie-Injection XSS

Correct Answer: a, b, c

Detailed Solution: Stored, Reflected, and DOM-based XSS are the three classical, recognized categories of Cross-Site Scripting attacks. "SQL-based XSS" and "Cookie-Injection XSS" are not standard or recognized XSS classifications.

Thus, the correct options are (a), (b), and (c).

QUESTION 10:

Consider the following simplified query used by a login form:

SELECT * FROM users WHERE username = 'INPUT1' AND password = 'INPUT2'

Which of the following input(s) for the username field, if unsanitized, would represent a classic SQL injection attempt to bypass authentication?

a. admin
b. admin' OR '1'='1
c. admin123
d. 'OR 1=1 –
e. guest_user

Correct Answer: b, d

Detailed Solution: Both "admin' OR '1'='1" and "' OR 1=1 --" are classic SQL injection payloads that manipulate the query's logic to always evaluate as true, potentially bypassing authentication. The other options are normal, non-malicious input values.

Thus, the correct options are (b) and (d).

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